

Selection sort

- IDEA:

- Find the smallest element in the array and exchange it with the element in the first position, then find the second smallest element and exchange it with the element in the second position, and continue in this way until the entire array is sorted.

- Selection sort is:

- One of the simplest sorting techniques.
- Good algorithm to sort a small number of elements

Selection Sort



Algorithm Selection Sort (A [], n)

```
{
  For i = 0, i < n-1, i++)
  {
    int min=i;
    for(j=i+1, j < n, j++)
    {
      if(a[j] < a[min])
      {
        min=j;
      }
    }
    if(min != i)
    {
      swap(a[i], a[min])
    }
  }
}
```

Selection Sort - Complexity

Case	Time Complexity
Best Case	$O(n^2)$
Average Case	$O(n^2)$
Worst Case	$O(n^2)$